10/12/2026: Smallest Counterexample

CSCI 246: Discrete Structures

Textbook reference: Sec 21, Scheinerman

Today's Agenda

- ($\approx$ 30 min) Mini-lecture
- ($\approx$ 20 min) Group exercises

The Well-Ordering Principle

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This set is a nonempty subset of the natural numbers. By the well-ordering principle, S contains a least element: ???

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- Conclude: The claim holds!

Proof by smallest counterexample: A data science example

Setting. Data often arrives as a **stream** (sensor readings, web logs, a file too big for memory). We can't store everything and then average, so we keep a **running mean**, updating it as each point x_n arrives:

$$\mu_1 = x_1, \qquad \mu_n = \mu_{n-1} + \frac{x_n - \mu_{n-1}}{n} \quad \text{for } n \geq 2.$$

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Why care? We've proven a one-pass streaming algorithm correct for *every* input length. ✓

The most famous proof by smallest counterexample

Prime Factorization Theorem (existence part of the Fundamental Theorem of Arithmetic)

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Why should we care? Internet Security! Your bank account is safe because every large number has a prime factorization, but nobody can find it.

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- What does that mean?

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- What do we know about a, b?

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- So what?

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- Since s is the smallest element of X, neither a nor b is in X. So each is a product of primes.
- Then $s = ab$ is a product of primes, contradicting $s \in X$. ■

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Remark. This is the statement that "every positive integer has a binary representation." For example,

$$13 = 8 + 4 + 1 = 2^3 + 2^2 + 2^0 = 1101_{\text{two}}.$$

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Therefore X is empty, and the statement holds. ✓

Proof by the Well-Ordering Principle

Scheinerman's Proof Template 16.

Proof by the Well-Ordering Principle.

To prove a statement about natural numbers:

Proof. Suppose, for the sake of contradiction, that the statement is false. Let $X \subseteq \mathbb{N}$ be the set of counterexamples to the statement. (I like the letter X for eXceptions.) Since we have supposed the statement is false, $X \neq \emptyset$. By the Well-Ordering Principle, X contains a least element, x.

(Basis step.) We know that $x \neq 0$ because *show that the result holds for* 0*; this is usually easy.*

Consider $x - 1$. Since $x > 0$, we know that $x - 1 \in \mathbb{N}$ and the statement is true for $x - 1$ (because $x - 1 < x$). *From here we argue to a contradiction, often that x both is and is not a counterexample to the statement.*$\Rightarrow\Leftarrow$ ■

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Remarks.

- The basis step could be a different number than 0 (e.g. 1) if the statement is about a subset of the natural numbers (e.g. the positive integers).
- The contradiction needn't come from considering $x - 1$.

Group exercises